We have so far seen a few experimental results that needed a radically new picture of light to explain them. In particular, light cannot behave simply as a classical electromagnetic wave, but rather it also behaves as if it is composed of quantized particles called photons, which carry energy and momentum:

$$E = hf \quad \text{and} \quad p = \frac{h}{\lambda} \qquad \hbar = \frac{h}{2\pi}$$

$$\text{or: } E = \hbar\omega \quad \text{and} \quad \vec{p} = \hbar\vec{k}$$

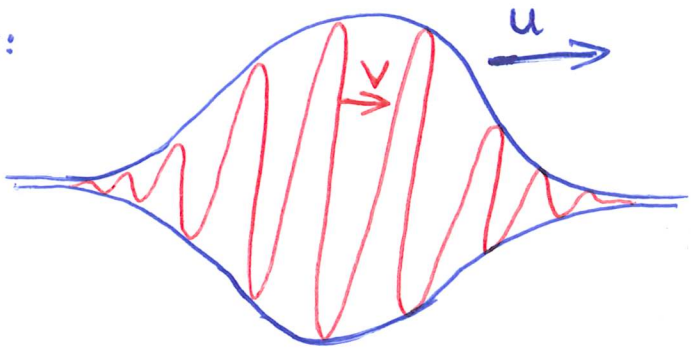
In his PhD thesis in 1924, Louis de Broglie (pronounced "de broy") proposed that if light behaves as both a wave and a particle, then other known particles should also behave like waves. This wave-like nature of matter has been come to be known as de Broglie matter waves, and they are an integral part of the quantum description of matter. de Broglie's PhD thesis was a single page!

His idea is that if a photon's energy is related to its frequency $E=hf$ and its momentum is related to its wavelength $p=h/\lambda$ then since a particle such as an electron has an energy and a momentum, it should have a corresponding frequency and wavelength (or equivalently ω and k)

In general, there are 2 wave velocities to worry about:

- phase velocity ⇒ velocity at which the phase of the wave moves
- group velocity ⇒ velocity at which energy and momentum is carried by the wave

wave packet:



the phase can move faster than the wave packet's envelope, but this is not the speed the wave moves at, this is the phase velocity.

The wave packet envelope moves at the group velocity u and is therefore the speed the particle moves at.

de Broglie: $E=\hbar\omega$ and $p=\hbar k$

relativity: $E=\gamma mc^2$ and $p=\gamma mu$ where $\gamma=\frac{1}{\sqrt{1-u^2/c^2}}$

Starting from $\lambda f=\frac{\omega}{k}$ we can find the dispersion relation for matter waves:

$$\lambda f=\frac{\omega}{k}=\frac{E/\hbar}{p/\hbar}=\frac{E}{p}=\frac{\gamma mc^2}{\gamma mu}=\frac{c}{\beta} \quad \text{where } \beta\equiv\frac{u}{c}$$

$$\Rightarrow \boxed{\frac{\omega}{k}=\frac{c}{\beta}}$$

Note that for photons $u=c$ so $\beta=1$ and the normal dispersion relation is recovered.

We will only consider non-relativistic matter in this course, so $E = mc^2$ and $p = mu$ $\left(\gamma = \frac{1}{\sqrt{1-u^2/c^2}} \approx 1\right)$

Example: What is the de Broglie wavelength of a golf ball travelling at 200 km/h? The mass of a golf ball is 46 g and its diameter is 42.7 mm.

Solution: The golf ball will have a momentum

$$p = mu = (0.046\,\text{kg})(200\,\text{km/h})\left(\frac{1000\,\text{m/km}}{3600\,\text{s/h}}\right) = 2.56\,\text{kg}\cdot\text{m/s}$$

The momentum is related to the matter wave's wavelength

$$p = \frac{h}{\lambda} \Rightarrow \lambda = \frac{(6.626\times10^{-34}\,\text{J}\cdot\text{s})}{2.56\,\text{kg}\cdot\text{m/s}} = \boxed{2.59\times10^{-34}\,\text{m}}$$

This is 32 orders of magnitude smaller than the diameter of the golf ball, which is why we never see a golfball exhibiting wavelike properties on the golf course.

Example: What is the de Broglie wavelength of an electron (mass $m_e = 9.11\times10^{-31}\,\text{kg}$) moving non-relativistically with a kinetic energy $K = 1.2\,\text{keV}$?

Solution: instead of using $K = \frac{1}{2}m_e u^2$ to solve for u to then calculate $p = m_e u$, note that K is equivalently given by

$$K = \frac{p^2}{2m_e} \Rightarrow p = \sqrt{2Km_e}$$

Since K is given in keV, we can also write $m_e = \frac{m_e c^2}{c^2}$

because $m_e c^2 = 511\,\text{keV} \equiv E_0$

$$\Rightarrow p = \frac{\sqrt{2KE_0}}{c} \quad \text{and} \quad p = \frac{h}{\lambda} \quad \text{where } h = 4.136\times10^{-15}\,\text{eV·s}$$

Therefore $$\lambda = \frac{hc}{\sqrt{2KE_0}} = \frac{(4.136\times10^{-15}\,\text{eV·s})(2.998\times10^{8}\,\text{m/s})}{\sqrt{2(1.2\times10^{3}\,\text{eV})(511\times10^{3}\,\text{eV})}}$$

$$\lambda = 3.54\times10^{-11}\,\text{m} \quad \text{or} \quad \boxed{\lambda = 35.4\,\text{pm}}$$

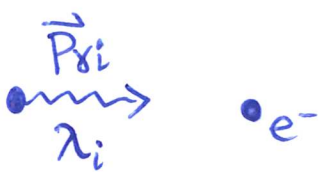
Note: this electron energy is on the low end of energies used in electron microscopy and even at this low energy the resolution achieved is ~4 orders of magnitude higher than with an optical microscope with wavelength ~400 nm.

5-minute break

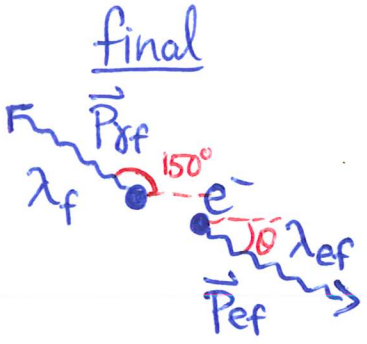
Let's look at a harder example making use of Compton scattering too:

Example: An x-ray photon of wavelength 18.0pm scatters off an electron at rest at an angle of 150°. What is the de Broglie wavelength of the electron after the collision?

initial



final



Solution: we need conservation of momentum but we also can find λ_f from Compton scattering.

$$\lambda_f = \lambda_i + (2.43\text{pm})(1-\cos 150^\circ) = 22.53\text{ pm}$$ final photon wavelength

For use in conservation of momentum:

$$\vec{P}_{\gamma i} = \frac{h}{\lambda_i}\hat{x} = \left(\frac{4.136\times10^{-15}\text{ eV}\cdot\text{s}}{18\times10^{-12}\text{ m}}\right)\hat{x} = (2.298\times10^{-4}\text{ eV}\cdot\text{s/m})\hat{x}$$

$$\vec{P}_{\gamma f} = \frac{h}{\lambda_f}\left(\cos 150^\circ \hat{x} + \sin 150^\circ \hat{y}\right) = 1.836\times10^{-4}\text{ eV}\cdot\text{s/m}\left(-\frac{\sqrt{3}}{2}\hat{x} + \frac{1}{2}\hat{y}\right)$$

$$\vec{P}_{ef} = P_{efx}\hat{x} + P_{efy}\hat{y}$$

x-component conservation:

$$P_{\gamma i} = P_{\gamma f}\cos 150^\circ + P_{efx} \Rightarrow P_{efx} = P_{\gamma i} - P_{\gamma f}\cos 150^\circ$$
$$= 3.888\times10^{-4}\text{ eV}\cdot\text{s/m}$$

y-component conservation:

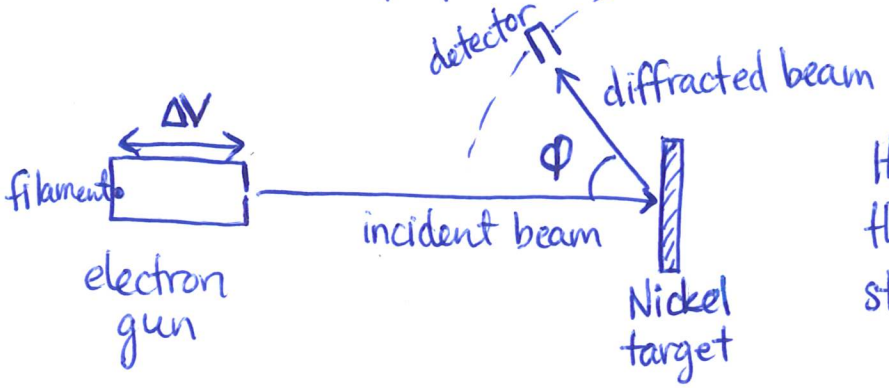
$$0 = P_{\gamma f}\sin 150^\circ + P_{efy} \Rightarrow P_{efy} = -P_{\gamma f}\sin 150^\circ$$
$$= -9.18\times10^{-5}\text{ eV}\cdot\text{s/m}$$

$\Rightarrow$ final electron momentum: $P_{ef} = \sqrt{P_{efx}^2 + P_{efy}^2}$
$$= 3.995\times10^{-4}\text{ eV}\cdot\text{s/m}$$

Now find $\lambda_{ef} = \frac{h}{P_{ef}} = \frac{4.136\times10^{-15}\text{ eV}\cdot\text{s}}{3.995\times10^{-4}\text{ eV}\cdot\text{s/m}} = 1.035\times10^{-11}\text{ m}$

$\Rightarrow$ the scattered electron has a de Broglie wavelength of $\boxed{10.35\text{pm}}$

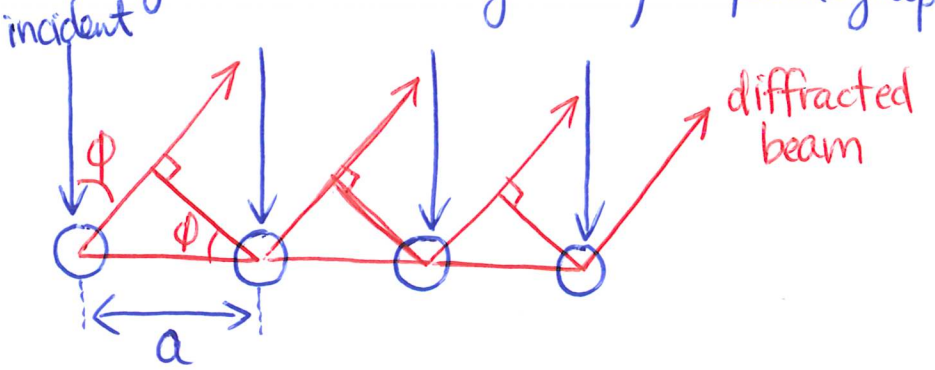
Experimental verification of de Broglie matter waves came in 1927 in the Davisson-Germer experiment confirming that electron beams exhibit wavelike properties.



Highly ordered Ni target so that the atoms form a regular crystal structure

Electrons are emitted by a hot filament and accelerated through a potential difference ΔV emerging as a highly collimated incident beam with kinetic energy $K = e\Delta V = \frac{p^2}{2m} \Rightarrow p = \sqrt{2me\Delta V}$.

The Ni target has a regular crystal lattice with lattice spacing independently measured using x-ray crystallography of $a = 2.15 \times 10^{-10}$ m.



There will be constructive interference if $a\sin\varphi = n\lambda$ $n = 1, 2, 3, \ldots$

At $\Delta V = 54$V the $n=1$ maximum was observed at scattering angle $\varphi = 50°$, which gives $\lambda = (2.15 \times 10^{-10}\text{m})\sin 50° = \boxed{1.65 \times 10^{-10}\text{m}}$. Meanwhile from the wave's momentum $p = \sqrt{2me\Delta V} = 2.478 \times 10^{-5}$ eV·s/m giving

$\lambda = \frac{h}{p} = \boxed{1.67 \times 10^{-10}\text{m}}$ in agreement with the diffraction prediction.